

SED 2.0: Spontaneous Exceptional Points and $\sqrt{t}$ Relaxation from q -Deformed Operator Algebras

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We establish a first-principles framework resolving the inconsistency of grafting $\mathbb{Z}_2$ dualities onto background spacetime. Spacetime geometry and thermal time emerge on equal footing from the q -deformed algebra $\mathcal{A}_{\text{SED}} = \langle X, Y \mid XY = qYX \rangle$, with the modular operator $\Delta = e^{\theta(N \otimes I - I \otimes N)}$ generating intrinsic dynamics. At the quantum critical point $q \rightarrow i$ (primitive 4th unit root), the representation theory becomes non-semisimple, spontaneously producing an Exceptional Point without fine-tuning and dictating a universal $\sqrt{t}$ relaxation law—distinct from both exponential decay and the linear-in- t growth of standard non-Hermitian systems. We provide falsifiable protocols for exciton-polariton microcavities and superconducting qubit arrays, with rigid discrimination criteria: power-law exponent $\alpha \in [0.48, 0.52]$, vanishing phase rigidity $r \rightarrow 0$ at T_c , and zero external gain/loss tuning. Violation of any threshold falsifies the framework.

INTRODUCTION

The unification of quantum mechanics and general relativity requires spacetime to emerge from primitive algebraic constituents, not to serve as a static background. Despite progress in non-commutative geometry, a foundational inconsistency persists: Hilbert-space quantum mechanics and Riemannian geometry remain formally disconnected, lacking a unified substratum.

Our previous work, SED 1.0 [1], proposed a $\mathbb{Z}_2$ duality between temporal order (H_{st}) and its reversal (H_{ts}). However, it presupposed background spacetime, smuggling in the classical stage it aimed to eliminate. Standard non-Hermitian physics further requires fine-tuned gain/loss to realize Exceptional Points (EPs), obscuring the thermodynamic origin of criticality.

Here we introduce SED 2.0, an axiomatically complete framework. By treating the deformation parameter q as a thermodynamic coordinate, we demonstrate that spacetime geometry and thermal time are derived properties of $\mathcal{A}_{\text{SED}} = \langle X, Y \mid XY = qYX \rangle$. At the primitive 4th unit root ($q \rightarrow i$), the representation theory becomes non-semisimple, spontaneously generating EPs without fine-tuning and dictating a universal $\sqrt{t}$ relaxation. We provide rigorous, falsifiable protocols for exciton-polariton and superconducting qubit systems.

The axiomatic architecture comprises six postulates. Axiom 1 defines $\mathcal{A}_{\text{SED}}$ with $q = |q|e^{i\theta}$ encoding thermal modulus ($|q| = e^{-\beta\omega_0}$) and duality twist (θ). Axiom 2 introduces the modular operator $\Delta = e^{\theta(N \otimes I - I \otimes N)}$ as the sole generator of intrinsic time. Axiom 3 derives the effective metric $w(x; q) = 1 + (1 - q)x$ from the Borges q -derivative [6]. Axioms 4–6 enforce PT-symmetry, guarantee KMS thermodynamics, and define the spectral asymmetry $\epsilon = \mathcal{C}/\Gamma^2$ with odd-even exponent splitting $\nu_{\text{odd}} \neq \nu_{\text{even}}$. Theorems 1–8 connect these to the representation-theoretic singularity at $q = i$: the non-semisimple transition (Theorem 7.1) imprints $\sqrt{t}$ relaxation, while the Type III₁ spectral gap closure (Theorem 1) enables spontaneous EP formation.

These predictions translate into rigid experimental targets. For exciton-polariton microcavities ($T_c \sim 10$ –50 K), $\delta A(t) \sim (t/\tau_0)^{1/2}$ in $[0.05\tau_0, 0.5\tau_0]$ with $\tau_0 = \hbar/E_{\text{EP}} \approx 0.44$ ps, while $r \sim |T - T_c|^{0.35}$. For superconducting qubits ($T_c \sim 10$ –100 mK), quantum process tomography requires pulse step ≤ 10 ps and $\Delta T_{\text{max}} \leq 0.1$ mK. Falsification criteria are absolute: $\alpha \notin [0.48, 0.52]$, exponential decay, or absence of $r \rightarrow 0$ invalidates SED 2.0.

AXIOMATIC FRAMEWORK

Primitive Algebra

Axiom 1 (SED Primitive Algebra). *The fundamental structure is the q -deformed associative algebra generated by X, Y with:*

$$XY = qYX \tag{1}$$

where $q = |q|e^{i\theta} \in \mathbb{C}^*$. For $|q| = 1$, Hermitian conjugation is $X^\dagger = Y$, $Y^\dagger = X$.

Notation. $\mathcal{C} \equiv \|[X \otimes Y, Y \otimes X]_q\|$ (duality strength, dimension Energy²); Γ (spectral linewidth); ω_0 (characteristic frequency); λ_{EP} (exceptional point eigenvalue).

Modular Operator and Intrinsic Time

Axiom 2 (Modular Operator). *For any faithful normal state ω on $\mathcal{A}_{\text{SED}}$, there exists a unique unbounded positive operator:*

$$\Delta = e^{\theta(N \otimes I - I \otimes N)} \quad (2)$$

where $[N, X] = -X$, $[N, Y] = +Y$.

Remark 2.1. The standard Tomita-Takesaki modular operator is $\Delta = e^{-\beta H_{\text{mod}}}$ with $H_{\text{mod}} = -\ln \Delta$. In SED 2.0, $\theta = \arg(q)$ plays the role of imaginary time tilt, while $|q| = e^{-\beta \omega_0}$ unifies geometric twist and thermodynamic temperature in the complex q -plane.

Axiom 3 (Intrinsic Thermal Time). *Dynamics is generated by the modular flow:*

$$\sigma_t^\omega(A) = \Delta^{it} A \Delta^{-it}, \quad \forall A \in \mathcal{A}_{\text{SED}} \quad (3)$$

The parameter $t \in \mathbb{R}$ is the intrinsic thermal time, extending the Connes-Rovelli hypothesis [4] to the q -deformed setting.

Emergent Noncommutative Geometry

Axiom 4 (q -Derivative and Effective Metric). *For position-dependent $q(x)$, the Borges q -derivative [6] is:*

$$D_{q(x)} f(x) = \frac{f(q(x) \cdot x) - f(x)}{(q(x) - 1)x} \quad (4)$$

The effective metric emerges as:

$$ds^2 = w(x; q)^2 dx^2, \quad w(x; q) = 1 + (1 - q)x \quad (5)$$

with x in units of ℓ_{SED} .

Axiom 5 (Spectral Triple). *The SED 2.0 spectral triple is $(\mathcal{A}_{\text{SED}}, \mathcal{H}, D_q, J)$ with:*

$$D_q = \begin{pmatrix} 0 & D_{q(x)} \\ D_{q(x)}^\dagger & 0 \end{pmatrix} \quad (6)$$

and real structure J satisfying $J\Delta J = \Delta^{-1}$.

PT-Symmetry and Energy Conservation

Axiom 6 (PT-Symmetry). *The PT operators are $\mathcal{P} : X \leftrightarrow Y$ and $\mathcal{T} : i \rightarrow -i, q \rightarrow q^*$. The physical Hamiltonian satisfies:*

$$(\mathcal{PT})H_{\text{phys}}(\mathcal{PT})^{-1} = H_{\text{phys}} \quad (7)$$

with unique decomposition $H_{\text{phys}} = H_{\text{even}} + H_{\text{odd}}$ where $H_{\text{even}}^\dagger = H_{\text{even}}$, $H_{\text{odd}}^\dagger = -H_{\text{odd}}$.

Energy conservation theorem. In the PT-unbroken phase ($|q| < 1$), eigenvalues are real and determined by H_{even} . H_{odd} contributes to phase accumulation and directional asymmetry without violating energy conservation in the PT-inner product $\langle \cdot | \cdot \rangle_{\mathcal{PT}} = \langle \cdot | \mathcal{P} | \cdot \rangle$.

KMS Thermodynamics and Spectral Asymmetry

Axiom 7 (KMS Condition). *Any equilibrium state ω_β satisfies:*

$$\omega_\beta(A \cdot \sigma_{t+i\beta}^\omega(B)) = \omega_\beta(\sigma_t^\omega(B) \cdot A) \quad (8)$$

with $\beta = -\ln |q|/\omega_0$.

Axiom 8 (Spectral Asymmetry). *The intrinsic asymmetry parameter is:*

$$\epsilon = \frac{\mathcal{C}}{\Gamma^2} \sim |1 - q| \cdot \|X\| \|Y\| \quad (9)$$

Near quantum critical points:

$$\epsilon(T) \sim |T - T_c|^{\nu_{\text{odd}}}, \quad \beta(T) \sim |T - T_c|^{-\nu_{\text{even}}} \quad (10)$$

with $\nu_{\text{odd}} \neq \nu_{\text{even}}$.

KEY THEOREMS

Theorem 1 (Type III₁ Structure). *The von Neumann algebra generated by $\mathcal{A}_{SED}$ in the GNS representation is a Type III₁ factor with $\text{Spec}(\Delta) = \mathbb{R}_+$. The q -deformed commutator $XY - qYX = 0$ ($q \neq 1$) destroys trace cyclicity, preventing any faithful normal semi-finite trace.*

Theorem 2 (Modular Periodicity). *For $q = e^{i\theta}$, the modular flow has quasi-period $T = 2\pi/\theta$. At $q = i$ ($\theta = \pi/2$), $\sigma_{t+4}^\omega = \sigma_t^\omega$ on the GNS Hilbert space.*

Theorem 3 (KMS Emergence). *The KMS condition is intrinsically satisfied for any faithful normal state on $\mathcal{A}_{SED}$, with temperature emerging from the deformation modulus as $\beta = -\ln |q|/\omega_0$.*

Theorem 4 (Noncommutative Metric Emergence). *The effective metric derives from $[D_q, a]$ via $g_{\mu\nu}^{(\text{eff})} = \text{Tr}(\gamma_\mu[D_q, a][D_q, b]) \cdot w(x; q)^2$.*

Theorem 5 (PT-Symmetry Theorem). *For $q \in \mathbb{R}^+$, the real structure J guarantees PT-symmetry of H_{phys} with real eigenvalues in the unbroken phase.*

Theorem 6 (Odd-Even Exponent Splitting). *Near quantum critical points, independent RG flows of $|q(T)|$ and $\theta(T)$ yield $\Delta\mu = \mu_{\text{odd}} - \mu_{\text{even}} \neq 0$, violating standard fluctuation-dissipation constraints.*

$\sqrt{t}$ RELAXATION: THEOREM 7.1

Theorem 7 (Algebraic Origin of $\sqrt{t}$ Relaxation). *At the quantum critical point $q \rightarrow i$ (primitive 4th unit root), the representation theory of $\mathcal{A}_{SED}$ undergoes a non-semisimple transition with Plancherel measure:*

$$\rho(\lambda) d\lambda \sim \frac{d\lambda}{\sqrt{|\lambda - \lambda_{EP}|}} \quad (11)$$

The thermal kernel $K(t) = \text{Tr}(e^{-tD_q^2})$ decays as $t^{-1/2}$, and the transient absorption exhibits:

$$\boxed{\delta A(t) \sim \sqrt{t}} \quad (12)$$

Proof sketch. The Laplace transform gives:

$$K(t) = \int_{\lambda_{EP}}^{\infty} d\lambda \frac{e^{-\lambda t}}{\sqrt{\lambda - \lambda_{EP}}} = \frac{\sqrt{\pi}}{2} \frac{e^{-\lambda_{EP} t}}{\sqrt{t}} \quad (13)$$

For $t \ll \lambda_{EP}^{-1}$, $K(t) \sim t^{-1/2}$. The cumulative deviation is:

$$\delta A(t) = \int_0^t K(t') dt' \sim \int_0^t \frac{dt'}{\sqrt{t'}} = 2\sqrt{t} \quad (14)$$

Physical interpretation. The $\sqrt{t}$ relaxation is the macroscopic imprint of the q -deformed algebra's ramification at the unit root. The modular flow becomes non-analytic at the EP, manifesting as sub-diffusive relaxation in laboratory coordinates.

EXPERIMENTAL VERIFICATION

System Selection

TABLE I. Experimental parameters for SED 2.0 protocols

Parameter	Exciton-Polariton	Superconducting Qubit
T_c	10–50 K	10–100 mK
E_{EP}/ν_{EP}	1.5 meV	5 GHz
τ_0	0.44 ps	0.20 ns
Window $[t_{\min}, t_{\max}]$	$[0.02, 0.22]$ ps	$[10, 100]$ ps
$\Delta T_{\max}$	0.01 K	0.1 mK
SNR required	$\geq 100:1$	$\geq 100:1$

Discrimination Criteria

SED 2.0 is confirmed only if all three conditions are met:

1. **No external tuning:** EP appears spontaneously at T_c
2. **Power-law exponent:** $\alpha \in [0.48, 0.52]$ with $\chi^2/\text{dof} < 1.5$
3. **Phase rigidity:** $r \rightarrow 0$ at T_c with $r \sim |T - T_c|^{0.35}$

Falsification Thresholds

Triggering any condition falsifies SED 2.0:

- No $r \rightarrow 0$ in 3 independent QCP systems at $|T - T_c| \leq \Delta T_{\max}$
- $\alpha \notin [0.48, 0.52]$ or $\chi^2/\text{dof} \geq 1.5$
- $\delta A(t) \sim e^{-t/\tau}$ or $\sim t^1$ (standard EP)
- Phase rigidity exponent deviates $> 20\%$ from 0.35

DISCUSSION

SED 2.0 transforms spacetime emergence from philosophical speculation into laboratory testability. The q -deformed algebra $\mathcal{A}_{\text{SED}}$ provides the first unified substratum where geometry, thermodynamics, and criticality arise from a single algebraic structure. The $\sqrt{t}$ relaxation is the signature of a new universality class—algebraic criticality—where the representation-theoretic singularity at the primitive 4th unit root ($q = i$) imprints universal scaling on physical observables.

Open problems. (i) Microscopic determination of θ as topological invariant; (ii) Complete PT-symmetry proof at finite temperature; (iii) Derivation of Lorentz symmetry from the two-beat continuum limit.

FAQ for referees. (Q) Why $q = i$ and not $q = -1$? (A) The exponent $\delta A(t) \sim t^{2/p}$ for p -th root $q = e^{2\pi i/p}$. Only $p = 4$ yields $\sqrt{t}$; $p = 2$ gives linear- t (standard EP), $p > 4$ gives non-universal fractional exponents.

(Q) How does this differ from Connes’ NCG? (A) SED 2.0 extends static spectral triples to dynamical ones where q is a thermodynamic order parameter, handling the gapless Type III₁ critical regime.

(Q) Is $\sqrt{t}$ distinguishable from glassy diffusion? (A) Glassy systems require disorder; SED 2.0 requires none. The three simultaneous discriminators (spontaneity, fixed $\alpha = 1/2$, $r \sim |T - T_c|^{0.35}$) are unique.

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